



Deep Tournament Selection for Genetic Algorithms

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*Equal contribution

1. MOTIVATION

- Selection drives the **exploration–exploitation** balance in GAs.
- Existing selection rules are mostly **static** and **handcrafted**.
- There are learning-based crossover and mutation operators. Selection has largely been left behind.

2. CORE IDEA

- Learn a **probability distribution** over tournament candidates.
- Train tournament selection online as a **Markov Decision Process (MDP)**, with **no extra fitness evaluations**.
- Use **Transformer self-attention** with global and local ranks to adapt **selection pressure**.

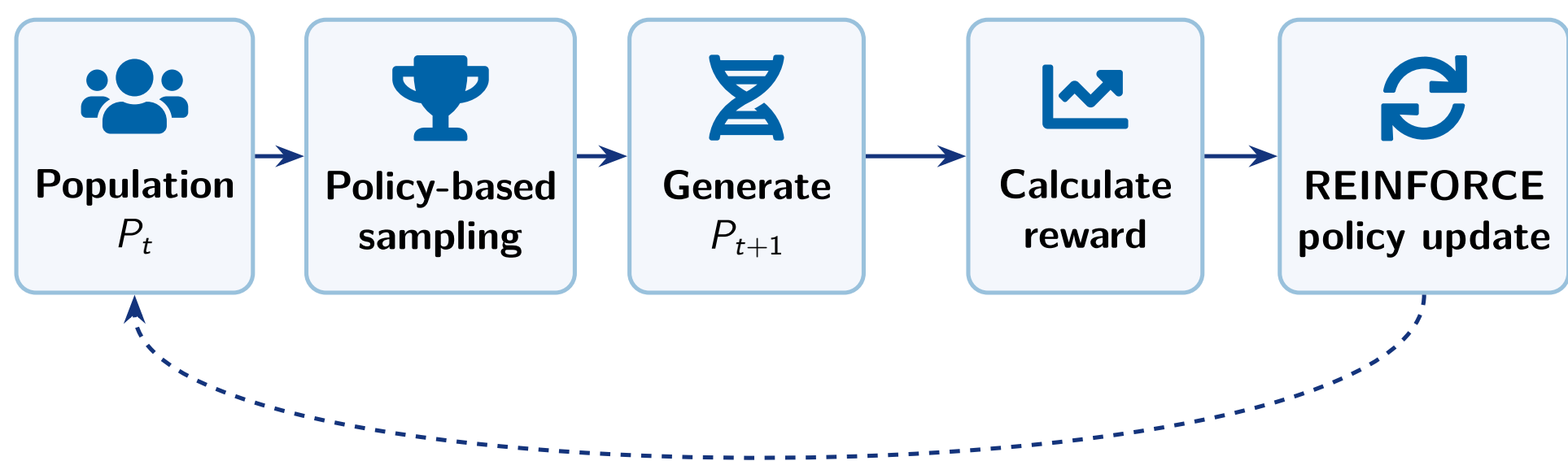
3. CONTRIBUTIONS

- Introduce **Deep Tournament Selection (DTS)**, a domain-independent operator that learns tournament-selection decisions.
- Propose a **Contextual Tournament Evaluation** architecture combining Transformer embeddings, global and local rank encodings, and a self-attention pointer mechanism.
- Design a **fully online training strategy** that interleaves optimization with the GA loop. We empirically demonstrate that this adapts selection pressure to achieve **faster convergence** and **better final quality** across Graph Coloring, Set Cover, and TSP with **negligible overhead**.

5. HOW DTS LEARNS

DTS treats tournament selection as a **Markov Decision Process** and learns the selection policy online using **REINFORCE**.

Online policy optimization



MDP formulation

- State:** Current population P_t with fitness values.
- Action:** Select one individual from each tournament.
- Policy:** Deep selection model (see **Method Overview** for more details).
- Reward:** Improvement in the average top-5 fitness between consecutive generations:

$$R_t = \frac{1}{m} \left(\sum_{f \in f_{t+1}^{(m)}} f - \sum_{f \in f_t^{(m)}} f \right), \quad m = 5$$

Online training

- Policy updates are performed online using **REINFORCE**.
- Uses fitness values already produced during evolution, requiring **no additional fitness evaluations**.
- Model updates are performed every **10 generations**.

6. EXPERIMENTAL SETUP

Benchmark domains

- Graph Coloring:** Minimize the number of colors while ensuring adjacent vertices receive different colors.
- Set Cover:** Minimize selected subsets while ensuring all universe elements are covered.
- Traveling Salesman Problem (TSP):** Minimize tour length over valid city permutations.

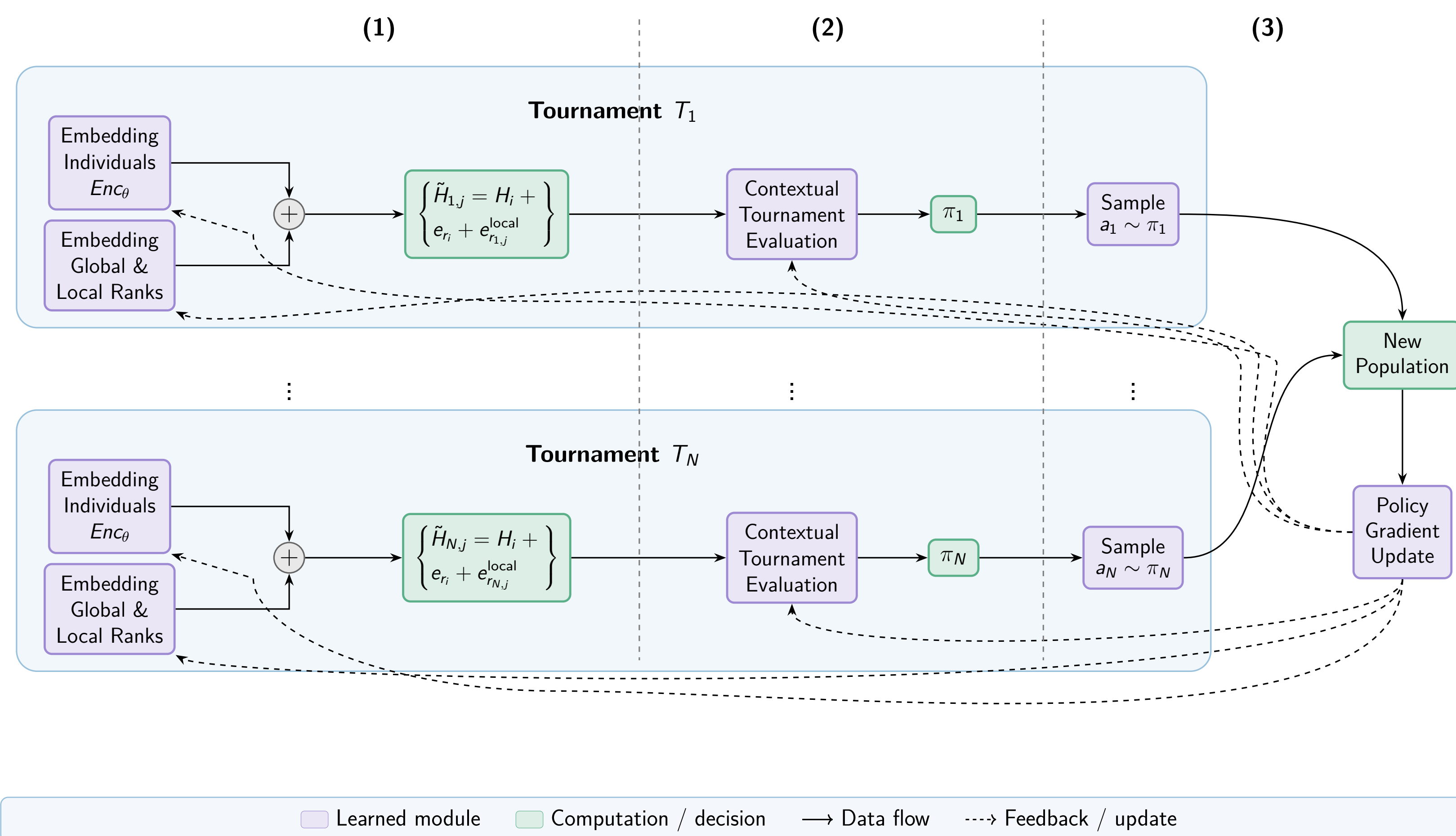
Baseline selection operators

- Fitness-based:** Roulette selection; Stochastic Universal Sampling (SUS).
- Rank-based:** Linear rank; Truncation.
- Competition-based:** Tournament.
- Adaptive:** Selection operator decider genetic algorithm (SODGA).

Comparison protocol

- Evaluation:** Each baseline selection operator is run on all benchmark problems, with 15 runs per instance; statistical significance is assessed.
- Fairness comparison:** Identical GA parameters and budgets are used, with the selection operator being the only variation.

4. METHOD OVERVIEW



(1) Embed each individual and add global & local rank encodings.

(2) Evaluate the tournament jointly to obtain a selection policy π .

(3) Sample a winner; elite-fitness gain rewards an online policy update.

7. RESULTS

1. Final solution quality* (average over 15 runs; lower is better)

	Instance	SUS	Linear Rank	Truncation	Tournament	SODGA	DTS (ours)
Graph Coloring	games120	41.0 (1.7)	43.2 (2.3)	28.6 (1.2)	34.7 (2.2)	33.7 (2.3)	25.0 (2.2)
	myciel7	76.3 (2.4)	82.7 (3.2)	59.7 (2.0)	64.1 (2.4)	63.1 (2.1)	46.3 (5.4)
	zero1n.i.2	92.3 (3.1)	97.6 (2.7)	73.1 (2.5)	75.0 (2.6)	74.0 (2.7)	64.0 (2.6)
Set Cover	scp41	301.2 (23.2)	257.2 (11.6)	176.7 (29.2)	74.2 (3.7)	87.3 (4.9)	67.5 (2.8)
	scp51	657.7 (45.9)	597.6 (14.4)	353.0 (63.3)	130.0 (9.4)	125.8 (6.4)	99.6 (9.5)
	scp65	265.5 (30.6)	223.0 (9.3)	98.4 (19.1)	43.5 (2.5)	41.3 (2.2)	38.6 (1.7)
TSP	berlin52	7.66 (0.13)	7.78 (0.20)	8.00 (0.19)	8.00 (0.18)	7.76 (0.10)	7.56 (0.04)
	lin105	14.89 (0.07)	15.26 (0.34)	15.66 (0.69)	15.22 (0.35)	15.23 (0.29)	14.65 (0.07)
	att48	10.78 (0.14)	10.81 (0.11)	11.19 (0.24)	10.96 (0.25)	10.76 (0.09)	10.70 (0.05)

*Roulette Wheel selection underperformed consistently across all domains and was omitted for brevity.

DTS achieves the best or tied-best final objective on every benchmark instance

2. Mean wall-clock time per generation (averaged over all instances; seconds)

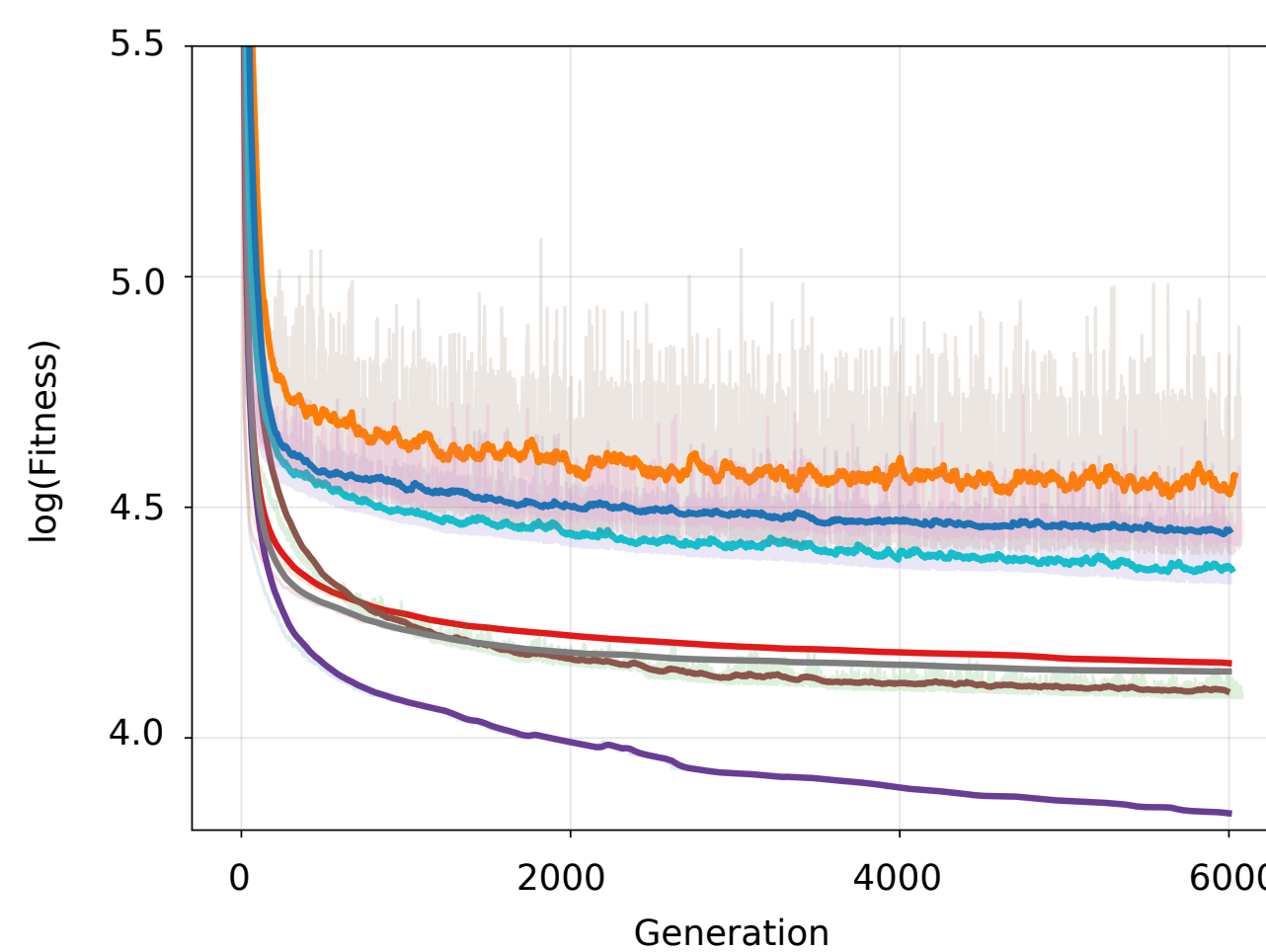
Domain	Roulette Wheel	SUS	Linear Rank	Truncation	Tournament	SODGA	DTS (ours)
Graph Coloring	0.05	0.05	0.05	0.05	0.04	0.03	0.05
Set Cover	0.25	0.25	0.25	0.25	0.19	0.15	0.20
TSP	0.02	0.01	0.02	0.02	0.02	0.02	0.02

DTS adds negligible runtime overhead and remains comparable to standard tournament selection

Representative instance: myciel7 (Graph Coloring)

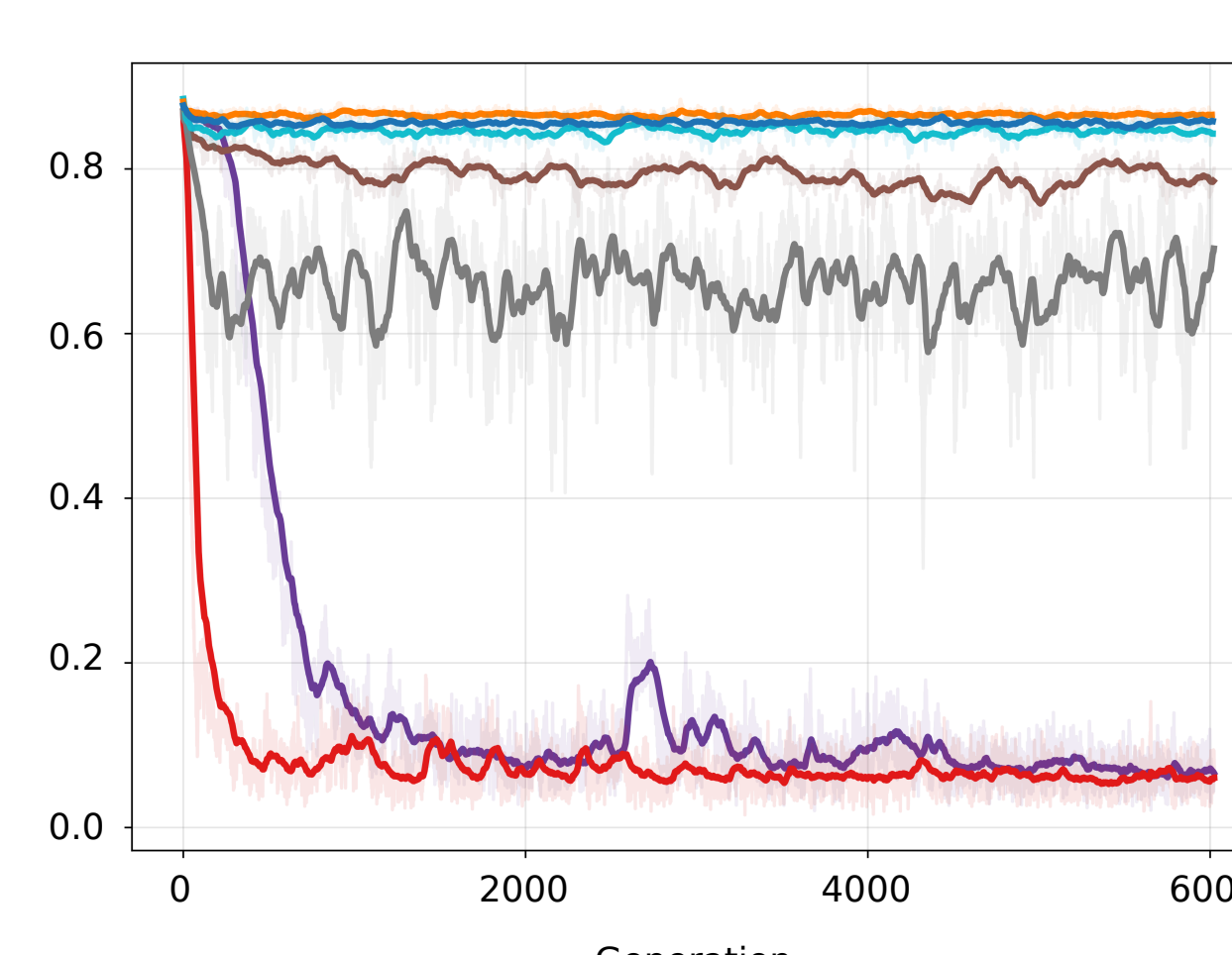
3. Convergence

(best individual per generation)



4. Population Diversity

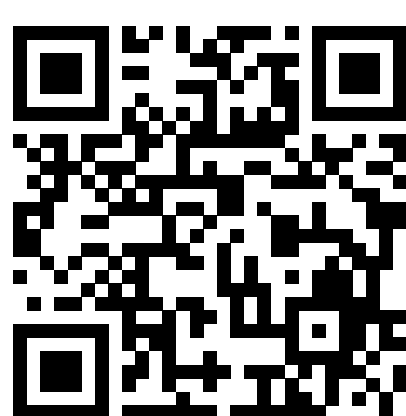
(average per generation)



DTS reaches lower fitness earlier and keeps improving

DTS maintains higher diversity than tournament

SCAN AND CONTACT



Paper & Repo QR

github.com/EC-KitY/DTS-for-GA
Paper, code, figures, and data.

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8. CONCLUSIONS

DTS learns adaptive selection pressure, improving quality, convergence, and diversity



Evidence

DTS improves quality and convergence across benchmarks.



Takeaway

Selection can be learned **online** from the evolving population.



Mechanism

Fixed selection pressure becomes a **context-aware policy**.



Next

Extend learned selection to richer encodings and GP.